

DBMS QUESTION BANK(C2311C4)

- 1. List and explain the advantages of DBMS over file based system. OR Explain disadvantages of files based system.**
- 2. Draw and explain 3 level architecture of DBMS.**
- 3. Compare: DDL and DML**
- 4. What is the difference between DELETE and TRUNCATE commands?**
- 5. List and explain different categories/types of database users.**
- 6. List and explain different tasks/roles/functions/duties of DBA (Database Administrator).**
- 7. Explain DBMS architecture with block diagram. OR Explain Database System architecture with block diagram.**
- 8. Write a note on mapping cardinality in E-R diagram.**
- 9. Explain the difference between a weak and a strong entity set.**
- 10. Explain the difference between generalization and specialization. OR Explain specialization and generalization concept in E-R diagram with suitable example.**
- 11. Write a note on constraints on specialization and generalization.**
- 12. Explain aggregation in E-R diagram with example.**
- 13. What do you mean by integrity constraints? Discuss various integrity constraints.**
- 14. Explain Different Data model in Database.**
- 15. Draw E-R diagram for Bank Management System.**
- 16. Define E-R diagram. Draw an E-R diagram for Library Management System. Assume relevant entities and attributes for the given system.**
- 17. Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.**

18. Design a generalization–specialization hierarchy for a motor-vehicle sales company. The company sells motorcycles, passenger cars, vans, and buses. Justify your placement of attributes at each level of the hierarchy. Explain why they should not be placed at a higher or lower level
19. Describe the DDL, DML, DCL commands for the student’s database, which contains student details: name, id, DOB, branch, DOJ, and course details: Course name, Course id, Stud Id, Faculty name, id, marks.
20. Define Super key, Primary key, Candidate key and Alternate key.
21. Explain Structure of Relational Databases.
22. Difference between open source and commercial database.
23. Explain following Relational Algebra Operation with example.(Refer all examples which we solve during lecture.)
- Selection
 - Projection
 - Cross Product
 - Joins (Inner Join, Outer Joins)
 - Rename
 - Division
 - Set operators
24. Explain different aggregate functions with example.
25. Explain Natural join or inner join and outer join with example.
26. Consider the following relational database, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries
- employee (ssn, name, dno, salary, hobby, gender)
 - department (dno, dname, budget, location, mgrssn)
 - works_on (ssn, pno)
 - project (pno, pname, budget, location, goal)
- i. List all pairs of employee names and the project numbers they work on.

- ii. List out department number, department name and department budget.
- iii. List all projects that Raj Yadav works on by project name.
- iv. List the names of employees who supervise themselves.

20. Consider the following relational database, where the primary keys are underlined. Give an expression in the relational algebra to express each of the following queries

- course (course-id, title, dept_name, credits)
- instructor (id, name, dept_name, salary)
- section (course-id, sec-id, semester, year, building, room_no, time_slot_id)
- teaches (id, course-id, sec-id, semester, year)

- II. Find the name of all instructors in the physics department.
- III. Find all the courses taught in the fall 2009 semester but not in Spring semester.
- IV. Find the names of all instructors in the Comp. Sci. department together with the course titles of all the courses that the instructors teach.
- V. Find the average salary in each department.

1. What is meant by normalization? Write its need. List and discuss various normalization forms.

2. Consider schema EMPLOYEE(E-ID,E-NAME,E-CITY,E-STATE) and

FD = {E-ID → E-NAME, E-ID → E-CITY, E-ID → E-STATE, E-CITY → E-STATE}

- Find attribute closure for: (E-ID)⁺

3. Compute the closure of the following set F of functional dependencies for relation schema R(A, B, C, D, E).

F = { A → BC, CD → E, B → D, E → A }

- List the candidate keys for R.

4. Consider schema $R = (A, B, C, G, H, I)$ and the set F of functional dependencies $\{A \rightarrow B, A \rightarrow C, CG \rightarrow H, CG \rightarrow I, B \rightarrow H\}$. (Use F^+)
- Prove that $AG \rightarrow I$ Holds.
27. Explain each normalization form with example.
28. Describe in detail about the following (i) Non loss decomposition. (ii) Lossy decomposition.
29. Explain functional Dependency with example and its type.
30. What is closure of FD and attribute set? Explain with example.
31. What is canonical cover? explain with example.
32. Explain anomaly and its type.
33. Explain query processing steps. OR Discuss various steps of query processing with proper diagram
34. Explain Heuristics in optimization.
35. Explain the purpose of sorting with example with reference to query optimization.
36. Explain the measures of finding out the cost of a query in query processing.
37. What is transaction ? explain each state with diagram.
38. Explain Schedule with its type.
39. Explain Serializability with its type./ List the types of serializability.
40. List the phases of two-phase locking protocol
41. Explain Concurrency with its problem.
42. Explain Authorization and access control in brief.
43. What is the difference between data security and data integrity?
44. What is constraint? Explain types of constraints.
45. Write a PL/SQL block to print the sum of Numbers from 1 to 100.
46. Write a PL/SQL block to print the given number is prime or not.
47. Write a PL/SQL program for inserting even numbers in EVEN table and odd number in ODD table from number 1 to 50.
48. Explain Cursor in PL/SQL.
49. Explain View in PL/SQL.
50. Explain stored procedure with proper example.
51. What are triggers? Explain the advantages and the needs of triggers.

Note:- This question bank is prepared for reference purposes only. Students are advised to also refer to class examples, lecture notes, and practical exercises for a complete understanding of the subject.